

Thin-layer chromatography (TLC) was used to determine the best solvent system for the mobile phase to purify a mixture by column chromatography. The TLC results using different mobile phase solvent systems are given in the following table.

| Mobile phase              | TLC results                     |
|---------------------------|---------------------------------|
| 100% Hexanes              | One spot near the origin        |
| 100% Ethyl acetate        | One spot near the solvent front |
| 1:1 Hexanes/ethyl acetate | Two spots separated             |

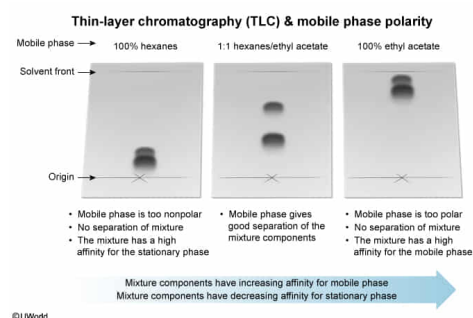
Which statement accurately describes the results?

- ☐ A. 100% Hexanes are too nonpolar because the mixture components have a higher affinity for the mobile phase. [8%]
- ✓ ☒ B. The 1:1 hexanes/ethyl acetate solvent system decreased the compounds' affinity for the mobile phase relative to 100% ethyl acetate. [65%]
- ☐ C. 100% Ethyl acetate is too polar because the mixture components have a higher affinity for the stationary phase. [9%]
- ☐ D. The 1:1 hexanes/ethyl acetate solvent system increased the compounds' affinity for the stationary phase relative to 100% hexanes. [16%]

Correct

65% answered correctly

### Explanation:



**Thin-layer chromatography (TLC)** is a technique used to **separate** compounds based on **polarity**. In normal-phase TLC, the stationary phase is made up of a polar adsorbent material, typically silica ( $\text{SiO}_2$ ), and the mobile phase is an organic solvent that travels up the stationary phase via **capillary action**.

The **rate** at which a compound **travels up the plate** is a function of the **relative polarities** of the **compound** and the **solvent**. If a mixture remains near the plate's origin, a more **polar solvent** is needed to increase the compound's affinity for the **mobile phase** relative to its affinity for the **stationary phase**. Conversely, if a mixture travels with the solvent front, a more **nonpolar solvent** is needed to *decrease* the compound's affinity for the mobile phase relative to its affinity for the stationary phase.

The 1:1 hexanes/ethyl acetate solvent system is more polar than 100% hexanes and less polar than 100% ethyl acetate, and it achieved the compound separation essential for column chromatography. This solvent system **decreased** the mixture components' **affinity for the mobile phase** relative to 100% ethyl acetate.

**(Choice A)** When using 100% hexanes, the compounds did not separate and remained near the origin. Therefore, 100% hexanes are too nonpolar and result in the mixture components having a higher affinity for the *stationary* phase than the mobile phase.

**(Choice C)** When 100% ethyl acetate was used alone, both compounds moved with the solvent front. Therefore, ethyl acetate is too polar of a solvent and results in the mixture components having a higher

affinity for the *mobile* phase than for the stationary phase.

**(Choice D)** The 1:1 hexanes/ethyl acetate solvent system increased the mixture components' affinity for the *mobile* phase, rather than the stationary phase, relative to 100% hexanes.

**Educational objective:**

Thin-layer chromatography is a technique used to separate compounds based on polarity. The rate at which a compound travels up the plate is a function of the compound and solvent polarities. Nonpolar solvents decrease a compound's affinity for the mobile phase whereas polar solvents increase a compound's affinity for the mobile phase.

**Time spent:**0 sec

**Subject:**  
Organic Chemistry

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**System:**  
Separation Techniques, Spectroscopy, and Analytical Methods